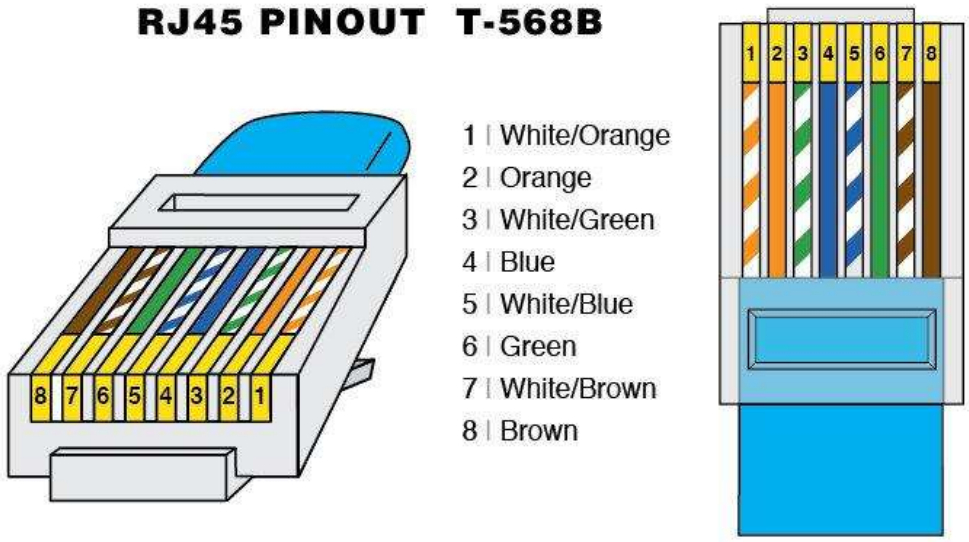
Network Cable And Connector

1.Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

ANS:

A Cat6 Ethernet cable is used to connect computers, routers and switches to a network.



Pin Number	Wire Colour
1	White-Orange
2	Orange
3	White-Green
4	Blue
5	White-Blue
6	Green
7	White-Brown
8	Brown

Pin Number and Wire Colours

Main Parts

- Cat6 Cable: Carries network data.
- RJ-45 Connector: Connects the cable to a computer or router.
- 8 Wires: Carry data signals.
- Pins 1–8: Show the wire positions.

2.List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

ANS:

Cable Type	Real-World Device / Example	Why It Is Used
1. Cat5e	Wi-Fi Router	Connects a router to a computer and provides internet.
2. Cat6	Zomato POS Machine	Provides fast internet for taking orders and making payments.
3. Fiber Optic	IPL Streaming Server	Provides very fast internet for live video streaming.

1. Cat5e: Used for normal internet connections.
2. Cat6: Used for faster network connections.
3. Fiber Optic: Used for very fast data transfer over long distances.

3.Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

ANS:

Step-by-step process:

1. Take a Cat6 Ethernet cable.
2. Remove the outer cover of the cable using a wire stripper.
3. Separate the 8 small wires.

4. Arrange the wires in T568B colour order.
5. Cut the wires evenly.
6. Insert all 8 wires into the RJ-45 connector.
7. Use a crimping tool to press the connector.
8. Test the cable using a cable tester.

Two Common Mistakes to Avoid

1. Wrong wire order: Always follow the T568B colour order.
2. Loose wires: Push all wires properly inside the connector before crimping.

Real-world use: RJ-45 cables connect computers, Wi-Fi routers and switches.

4.Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

ANS:

1. Cat6 Ethernet Cable: Connects computers, gaming consoles and routers. It provides fast and stable internet.
2. RJ-45 Connector: Used to connect the Cat6 cable to the router and devices.
3. HDMI Cable: Connects gaming consoles and computers to TV screens. It gives clear video and sound.
4. Fiber Optic Cable: Can connect the main network over long distances with high speed.

Real-World Example: During an IPL screening, Cat6 cables provide internet, and HDMI cables show the live match on big TV screens.

5.Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

ANS:

Ethernet Cable (RJ-45 Connector)



1. Cable Type: Cat6 Ethernet Cable.
2. Connector: RJ-45.
3. It connects a Wi-Fi router to a computer.
4. It is used to provide a fast internet connection.

Real-World Use: At home, an Ethernet cable connects a computer to a Wi-Fi router for internet access.